not described by equations of evolution in time *t*. What we must do is simply to enumerate the variables *A*, *B*, *C* … which we *actually* observe, and write equations expressing relations between *these* variables, and nothing else: that is, equations for the relations *A(B)*, *B(C)*, *C(A)* … which we observe, and not for the functions *A(t)*, *B(t)*, *C(t)* … which we *do not* observe.

In the example of the pulse and the candle chandelier, we will not have the pulse and the candelabrum evolving in time, but only equations which tell us how the two variables evolve with respect to each other. That is to say, equations which tell us directly how many pulse-beats there are in an oscillation, without mentioning *t*.

'Physics without time' is physics in which we speak only of the pulse and the chandelier, without mentioning time.

It's a simple change – but from a conceptual point of view, it's a huge leap. We must learn to think of the world not as something which changes in time but in some other way. Things change only in relation to one another. At a fundamental level, there is no time. Our sense of the common passage of time is only an approximation which is valid for our macroscopic scale. It derives from the fact that we perceive the world in a coarse-grained fashion.

The world described by the theory is thus far from the one we are familiar with. There is no longer space which *contains* the world, and no longer time *during the course of which* events occur. There are elementary processes in which the quanta of space and matter continuously interact with each other. Just as a calm and clear Alpine lake is made up of a rapid dance of a myriad of minuscule water molecules, the illusion of being surrounded by continuous space and time is the product of a long-sighted vision of a dense swarming of elementary processes.

Spacetime sushi

How do these general ideas apply to quantum gravity? How can we describe change without the ideas of space as a container, or time along which the world glides?

Consider a process: for example, the collision of two billiard balls on a table's green baize. Imagine a red ball played in the direction of a yellow one; it gets close, collides, and the two balls move away in different directions. This process, like all processes, takes place in a finite zone of space – let's say on a table approximately two metres wide – and lasts for a finite interval of time – let's say three seconds. To deal with this process in the context of quantum gravity, it is necessary to include space and time in the process itself (figure 7.1).

We must not, in other words, describe only the two balls, but also all that is around them: the table and any other material objects – and the space in